

Dynamic crowd evacuation approach for the emergency route planning problem: Application to case studies

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ABSTRACT

Disastrous situations, either natural or man-made, have claimed countless lives. In a planning context, this triggers the need for an evacuation situation that involves the emergency route planning (ERP) problem. ERP emphasizes the effectiveness of the evacuation plan to mitigate risk and ensure crowd safety. Reviews on previous ERP works have elicited a plethora of approaches, such as decision support, exact, heuristic, and meta-heuristic approaches. However, previous works on ERP focused on either crowd modeling techniques or routing approaches, but not both. This elicits the need to bridge modeling and routing approaches by means of a dynamic concept. To capture this dynamism concept, group cohesiveness and flexible routing approaches have been proposed, namely as an integrated evacuation planning with dynamism (iEvaP+). Group cohesiveness is utilized by considering group structure and intra-group compliance. Meanwhile, an artificial immune system (AIS) algorithm is adopted as the route optimizer. The iEvaP+ approach is applied to three case studies in order to identify the impact of crowd dynamism in the ERP problem. The result indicates that dynamism has direct influence on the evacuation plan performance. Besides the group cohesiveness, the properties of the evacuation network (such as exit) are also important in mitigating risks and crowd survivability.

1. Introduction

An emergency can be defined as a situation requiring time-critical response that is induced by a disastrous phenomena, be it naturally (i.e. hurricane, flood, earthquake) or man-made (i.e. hazard material release, fire), that could put human lives at risks (Alsnih and Stopher, 2004). When an emergency response is elicited, evacuation is the typical strategy employed to mitigate risks which require immediate mobilization and time-critical actions involving efficient coordination, space capacity utilization, and availability of emergency resources. In the context of planning, emergency evacuation is also known as emergency route planning (ERP) problem, encompassing two inter-related problems (Chiu et al., 2007): routing of the crowd and regulating the crowd flow.

Crowds can be formed by several or thousands of people that move in a bounded environment, either in group or individually, with respect to their individual goals in space such as avoiding obstacles, and remaining close to friends or family (Yersin et al., 2008). The definition of crowd is dependent on three co-related aspects (Sharma, 2009): (1) goals and needs; (2) social and physical attributes (e.g. level of interaction); and (3) psychological and situational aspects (stress levels at a respective time or place). The crowd model considered in this study is

tuned towards fulfilling the first two aspects, which is typically known as group cohesion.

Meanwhile, the perceived environment is utilized by crowds to choose the shortest path in time and space that leads to their goal (Yersin et al., 2008). The need for evacuation is crucial for residents in a bounded environment (within a structure or enclosed area) as opposed to those in an open space, due to the possible risks that are associated with it. As such, the attributes that can be associated with the bounded environment include the layout design (Shukla, 2009), its relation to risk factors (e.g. space capacity) (Park et al., 2007), and the available exit choices (e.g. exit's width) (Pu and Zlatanova, 2005).

In the context of crowd evacuation modeling and routing, dynamic concepts are important characteristics that aim to fulfill either level-of-service, design elements of pedestrian facilities, or planning guidelines (Helbing and Johansson, 2011). In a more specific term, a dynamic concept is an instance of synthetic human behavior when compared to actual human behavior (Fridman et al., 2011). In our study, the dynamic concept refers to the cohesiveness of groups in a crowd, particularly the grouping phenomenon (group formation) within the crowd and the level of interaction within the groups (intra-group compliance).

Crowds tend to propagate in a chaotic way during an emergency situation due to the uncertainty of the perceived situation (Wang et al.,

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2009). In addition, a crowd which moves in groups may affect the groups' flow of egression, their behavior toward others, and the estimated evacuation time (Wang et al., 2009). Additionally, environmental properties, such as exit choices, may have crucial impact on the life and death of people in the crowd during an emergency evacuation (Pu and Zlatanova, 2005). Understanding these underlying problems is crucial for crowd safety and can help emergency management agencies (EMAs) to prepare for and prevent an actual crisis.

Therefore, the objective of this study is twofold. Firstly, this study focuses on identifying the impact of the dynamic concepts (group cohesion) toward the crowd egression. Secondly, this study focuses on identifying the relationship between the exit properties (e.g. exit width and number) and the ability of the crowd egression. By achieving these objectives, the expected evacuation plan may help mitigate risks and ensure crowd safety prior to the occurrence of a disastrous situation.

2. Literature review

The cognition of the ERP approach involves three distinct and inter-related components: (1) crowd modeling, (2) route assignment, and (3) emergency's scenario modeling. However, our study is tuned towards fulfilling the first two aspects: crowd modeling and route assignment.

Crowd is generally modeled based on theoretical models, ranging from analytical ones to those based on matrices or cells (Bandini et al., 2005), generally categorized into microscopic, macroscopic, and mesoscopic (Radianti et al., 2013). Microscopic models treat every individual in the crowd as a separate "particle". Several variants of the microscopic approach include the encoding of human desires in the form of physics-based force, such as social force model (Helbing et al., 2000), and representing each pedestrian as a node that occupies a cell, such as cellular automata (Yuan and Tan, 2011). Macroscopic models describe crowds through their average flow and density. Fluid dynamic model (Helbing, 1998), flow tiles (Chenney, 2004), and non-local crowd dynamics (Colombo and Lécureux-Mercier, 2012) are variations of microscopic models. The hybrid of the two former models, namely the mesoscopic model, combines the key concepts of the relationship between local inter-individual interactions (micro) and collective patterns (macro) (Wang et al., 2009). Among these three categories of crowd models, the mesoscopic model is distinguished as the modeling approach that is able to capture crowd behavior realistically. Indirectly, adoption of this model helps in making an effective decision, improving response capability towards disaster, and reducing any adverse impacts on both human beings and surroundings (Lv et al., 2012).

The network or route is a logical layout which is used to model the perceived environments in time and space (Yersin et al., 2008). An actual building or structure layout is approximated by a continuous model. This is accomplished by means of networking a set of destination points that represents the real roadways for utilization by the crowd during emergency evacuation. Graph-based method is the most popular method adopted for approximating information of the enclosed region of a structure through a network of nodes and edges (Kemloh Wagoum et al., 2012). This is because the destination points can be pre-determined (i.e. exits) or adjustable (i.e. crossings, turning point at the end of a corridor). Also, the visibility of the graph of a minimal network at any location is ensured based on the range of at least one node. Consequentially, both logical approximations of the perceived environments and crowd modeling are interdependent and equally important in ERP. Unfortunately, research employing both these methods simultaneously is scarce.

Employing crowd modeling and graph-based layout representation, simultaneously, would help to efficiently move a crowd in a building. This is especially true in large and complex structures (Hajibabai et al., 2007), where actual agglomeration of the crowd is reflected. In a sociological sense, people who have social ties and intentionally walk together, such as family members and friends, are commonly known as "group" (Qiu and Hu, 2010; Moussaïd et al., 2010). In an emergency

context, the group characteristics (e.g. group size and influences among group members) may also influence the crowd egress flow as well as the efficiency of an evacuation plan (Qiu and Hu, 2010; Moussaïd et al., 2010; Moreland et al., 2013). This group characteristic is generally known as the group cohesion, which is defined as the tendency for a group to be in unity while working towards a goal or to fulfill the demands of its members (Carron and Brawley, 2000). A group's cohesive properties such as group size and influences among group members, are vital due to the following reasons: (1) The group size determines the group's physical structure and composition of the group; (2) Individual compliance influences the group's interaction in terms of their flow rate.

The dynamism concept has previously been adopted in the decision support system (DSS), where timely decision making before and during a disastrous situation is provided. Some examples include: Dynamic crowd navigation in a multi-storey building (Kwan and Lee, 2005); topological support and analysis of crowd distribution and simulation (Castle and Longley, 2005); route search and interactive evacuation instructions (Pu and Zlatanova, 2005); and dynamic route generation for both rescue and evacuation operations (Chang et al., 2009). Although the dynamism concept is captured, the focus of the studies is more on decision dynamics of the crowd egress route and not the dynamism of the crowd itself. In addition, DSS approaches are dependent on information availability, data security, and privacy compliances (Kwan and Lee, 2005).

ERP approaches that capture the dynamic concept using exact approaches have also been proposed. Examples of such approaches include: the panic reactions that attenuate the egress of a crowd (Wang et al., 2009); variable travel times and demand of crowd (Hui et al., 2010); crowd egress in reaction to their injury level (Amaldi et al., 2010); crowd with special needs (e.g. elderly and children) (Kaisar et al., 2012); network flow with reorganization of traffic routing during an evacuation (Bretschneider and Kimms, 2011); and dynamic crowd flow distribution during an evacuation (Lv et al., 2012). These approaches introduced dynamism in the form of crowd reactivity towards danger which impedes the overall evacuation process. Therefore, the dynamic nature of the approaches is limited towards the one that affects the formulated crowd model while not considering the dynamism of their route layout.

Meanwhile, heuristic is another popular and emerging ERP approach in the field of computational optimization. Examples of heuristic approaches are: capacity constrained route planner (CCRP) which dynamically utilizes spatial information for crowd routing (Lu et al., 2005; Kim et al., 2007; Zeng and Wang, 2009; Shekhar et al., 2012); an adaptive evacuation route algorithm (AERA) utilizing a dynamic network during the event of a flood (Liu et al., 2006); and an event-driven way-finding approach which combines the shortest and quickest path heuristics (Kemloh Wagoum et al., 2012). The dynamic nature of the heuristic approaches are bounded with the decision complex of the crowd when exposed to certain measurable levels of danger. This motivates the initiation of the crowd dynamic routing. Nevertheless, dynamic interaction between the crowd model and their respective routing mechanisms is limited such that the crowd model may influence the routing mechanism but not otherwise.

Studies employing the ERP approach by means of a meta-heuristic algorithm have been proposed by several researchers. Examples of the approaches include: a simulated annealing algorithm embedded in dynamic network loading of crowd egress (Cepolina, 2005); an ant colony algorithm to optimize the dynamic network flow of a crowd with respect to disaster extension (Yuan and Wang, 2007; Zong et al., 2010); and a tabu search for dynamic network optimization considering crossing and lane reversal strategies (Xie et al., 2010; Jiang et al., 2013). These approaches have demonstrated the dynamism of a crowd in network utilization, where adjustments of the crowd flow and changes in route selection are illustrated. Consequentially, the meta-heuristic approaches are limited towards achieving optimum

performance(s) of the evacuation procedure. Despite their stochastic nature, meta-heuristic approaches are able to reduce computational complexity (Yuan and Wang, 2007; Yusoff et al., 2008) and act as black boxes that do not take advantage of the problem (Talbi, 2009). Therefore, the meta-heuristic approach is deduced as ideal and suitable for solving the dynamic nature of the ERP problem.

Although various meta-heuristic approaches have been proposed for the ERP problems in the literatures, none of them have the features of artificial immune system (AIS) in their elementary structures. AIS can rapidly explore solution space while simultaneously exploiting the diversity of the solution, encapsulate the problems while preserving the solution's feasibility, and maintain elitism which eliminates the backward effect during iteration (Timmis et al., 2008). In addition, the AIS algorithm has been proven to solve a wide range of complex and combinatorial problems such as optimization, data mining, computer security, and robotics (Freschi and Repetto, 2005; Aickelin and Greensmith, 2007; Bagheri et al., 2010; Dasgupta et al., 2011). Despite AIS adoption in other areas of emergency management (e.g. logistic management (Chou et al., 2014)), its adoption to the ERP problem is scarce. Consequently, these features motivate the adoption of AIS algorithm as the proposed ERP approach.

While the dynamism concept has been adopted in some form, very little work had been done with regards to crowd dynamism focusing on the aspect of crowd grouping and dynamic routing of the crowd simultaneously. To the best of our knowledge, the work by Qiu and Hu (2010) is the first to introduce a unified framework of crowd grouping structure by a vector-based modelling approach, utilizing an agent-based simulation system, without dynamic routing of the crowd. Therefore, the motivation of this study is to propose a dynamic ERP approach which considers both crowd grouping and dynamic routing of the crowd simultaneously. In order to identify the impact of incorporating dynamism in the proposed ERP approach, it was applied to three case studies.

3. The proposed integrated evacuation planning with dynamism (iEvaP+)

The static crowd evacuation model can be found in the work by Khalid and Yusof (2014a) which defined the details of crowd grouping with full compliance levels. Also, the work on route selection optimization can also be found in their subsequent work (Khalid and Yusof, 2015). It defined the details of the adopted artificial immune system (AIS) algorithm as the route selection optimizer. The model and route optimizer is named the integrated evacuation planning (iEvaP) approach. Our study extends the iEvaP approach by the inclusion of dynamism concepts, thus naming it as integrated evacuation planning with dynamism (iEvaP+). The dynamic crowd evacuation model is discussed in Section 3.1 while the dynamic route selection optimizer is discussed in Section 3.2. The notations and parameters of the iEvaP+ approach used in this paper are identified and formalized (Table 1). Meanwhile, the performance measure of the study is to minimize the maximum network clearance time (NCT) (Eq. (1)).

$$\min NCT = \min \left\{ \max \left\{ \sum_{i=0}^{i=n} \sum_{j=1}^{j=n+1} N_{ij} = 0, \forall v_n \rightarrow V \right\} \right\}. \quad (1)$$

3.1. Crowd evacuation model with dynamism

The existing crowd models do not capture crowd grouping behaviors and the crowd intra-group relation simultaneously. This creates the need to incorporate the crowd model with group cohesion. The variation of group cohesion provides the foundation to observe different impact of group-based behaviors. This imitates actual world situations such as group structure and intra-group compliance level. Different group structures would enable variations of crowd flow, thus

Table 1
Notations and parameters.

Notation	
k	evacuees index
v	node index
a	edge index
g	group index
p	evacuation path with node sequences
M	total number of evacuees
V	set of nodes of the network
A	sets of edges of the network
η	a set of source/starting nodes
ζ	a set of exit/destination nodes
Parameter	
v_n	node index, where $n = 1, 2, \dots, n$
p_k	sequence number of node v_{pk}
d_g	group sizes
a_{ij}	edge from node v_i to node v_j
t_{ij}^k	time required for evacuee k to travel edge a_{ij}
C_n	maximum capacity of the node v_n
C_{ij}	maximum capacity to go through the edge a_{ij}
N_{ij}	number of evacuee k on edge a_{ij}

simulating group-based crowd egress movement. In addition, different compliance levels, such as cooperation between family members but not with strangers, may be subjected to the evacuees, as the occurrence of disaster tends to make them impatient (Wang et al., 2009). In corollary, by complimenting the concept of faster-is-slower effect (Wang et al., 2009), the interdependency and interaction among the crowds is also demonstrated using the variation of evacuee's compliance within a group.

In this particular setting, the group cohesiveness is quantified based on the group's size (d_g) and compliance level among group members ($rate_g$). Every group consists of different sizes (d_g), and each of the groups' starting points is different from each other. This cohesiveness is neglected if the evacuee does not belong to any group ($d_g = 1$). As stated by Carron and Brawley (2000), the minimum group size with measurable cohesiveness is $d_g \geq 2$, but for the maximum group size, it is dependent on the level of interaction and communication within them (Hackman, 2012). For simplicity, our study assumes the group size is between $2 \leq d_g \leq 9$ which is randomly assigned, whereas the ideal group size is $2 \leq d_g \leq 5$ (Carron and Brawley, 2000; Hackman, 2012; Moreland et al., 2013). Meanwhile, the group compliance level is randomly assigned between $10\% \leq rate_g \leq 100\%$ and is coupled with the d_g in order to simulate dynamic group cohesion. Any evacuee that belongs to a particular group, will be subjected to delay(s) based on the intra-group compliance level (Eq. (2)).

$$delay_g = \begin{cases} 1/2 * d_g * (rate_g/100) & \text{if } rate_g < 50\% \\ 1/4 * d_g * (rate_g/100) & \text{otherwise,} \end{cases} \quad (2)$$

$$g = 1, \dots, G$$

Based on Eq. (2), the delay caused by the variation of group sizes and group compliance levels are simulated. A situation with a higher compliance rate (cooperative) would results in lower competition among them but higher competition among different groups, thus improving flow rate, and vice versa. The overall evacuation plan is the resulting computation of $delay_g$, thus t_{ij}^k . The evacuation network's capacity is checked against the group sizes, where the allowance of passing through and possible delays are determined by two rules: (1) the travel time for the group remains the same t_{ij}^k , if $d_g \leq C_{ij}$, or (2) the travel time for the group is equal to the $(d_g/C_{ij}) * t_{ij}^k$, if $d_g > C_{ij}$.

3.2. Dynamic route optimizer using artificial immune algorithm (AIS)

In contrast to evolutionary algorithm, AIS solution instances are composed of populations of cells, in which, cloning is the main

mechanism of reproduction (which generates identical copies of cells). The operator for variation is the somatic hyper-mutation which performs mutations at a high rate on the cloned cells while network suppression is responsible for eliminating redundant solutions of the mutated cells (Bagheri et al., 2010; Dasgupta et al., 2011). In addition, AIS elementary structures naturally exploit solution diversity and rapid exploration of the search space, and always present solution elitism that eliminates backward effects during iteration (Freschi and Repetto, 2005; Aickelin and Greensmith, 2007). As such, these attractive AIS procedures are deemed to be suitable for adoption for solving the underlying problem.

The details of the adopted AIS algorithm is discussed in the work by Khalid and Yusof (2015). This section will highlight the dynamism concept of the AIS algorithm as the dynamic route optimizer. Two essential components of the dynamic route optimizer are dynamic routing and route optimization. The encoding scheme of the complete evacuation plan (population instance) of the proposed AIS algorithm is composed of arrays of candidate evacuation path p (antibody) matched with evacuee(s) (antigen); delineating the role of dynamic router. The evacuee may be an individual or within a group, which is situated at the source location η_{v_n} of the candidate p . The p may have subsets of v_{p_k} that are dynamically chosen by the evacuee k at certain time frames, but only by evacuees that started from the same η_{v_n} .

However, the capacities (C_n and C_{ij}) of the respective p are always enforced, thus avoiding an infeasible result. The flow rate of the crowd is simulated as a discrete event where each time frame is recorded as the NCT. The last evacuee egress time is the maximum NCT of the complete evacuation plan. The optimization mechanism works by reducing this maximum NCT by utilizing the altered operators of the proposed AIS. Finally, these core components (dynamic router and route optimization) defined the dynamic route optimizer of the iEvaP + approach (Fig. 1).

3.3. Data collection

Three case studies were considered for this research: Bukit Jambul shopping complex (BJ), Queensbay mall (QB), and the An-Nabawi mosque (NM). The architectural maps of BJ and QB were collected manually and represented as node-edge logical graphs which consist of nodes and edges. The data collected includes the number of nodes

which can possibly be approximated from the actual perceived environment, the possible source location and exit location which are accessible by public at all times, the travel time of designated edges, and the capacity of designated nodes. A similar approximation method was adopted for the third case study, where the area of An-Nabawi mosque was divided into 5 distinct regions. Region 1 (2 lower regions) is the region for the woman's praying area, while the remaining regions are the men's praying area. The details of the layout and region division can be referred to in Haron et al. (2012).

The BJ and QB case study data are collected from the ground floor area of their respective complexes, while the NM case study only involves the upper regions of the An-Nabawi mosque (Region 2, 3, and 4) due to computational limitations. All these case studies have been approximately represented by logical maps (Figs. 2–4), respectively. The travel time of the edges is computed by measuring the distance between one area to another, divided by the assumed flow velocity of 2 m/s for a pedestrian in an emergency situation (Helbing et al., 2003; Liu et al., 2009). This flow velocity is retained for the NM case study. However, to match the flow velocity of Malaysian citizens (BJ and QB case studies), the flow velocity of 20 persons are collected, where in average, 1.279 m/s is obtained. Therefore, the travel time of the edges of QB and BJ case studies data can be measured and obtained. Meanwhile, the capacity of ellipses and edges are computed by measuring the approximate areas (rooms/corridors/staircases) with the assumed average density of 0.45 person/m² (NM case study was scaled from the detailed layout and region division specified by Haron et al. (2012)). The travel time unit is in seconds.

The “N” of the source nodes (Figs. 2–4) represents anonymous numbers of evacuees, which is assigned with different values of M to imitate different levels of crowdedness of the respective case studies. As each source node may sustain a different number of evacuees, the distribution of the total number of evacuees M are varied based on the respective case study (Table 2), where the option numbers were designated for parameter optimization (refer to Section 4, Table 3). The BJ, QB, and NM case studies represent small (≤ 1000 evacuees), moderate ($\leq 10,000$ evacuees), and large ($\leq 50,000$ evacuees) sizes, respectively. The total capacity of the BJ, QB, and NM case studies are 982 evacuees, 11,924 evacuees, and 46,606 evacuees, respectively.

An actual evacuation plan can be complicated due to a variety of factors, constraints, uncertainties, and etc. The factors involved are

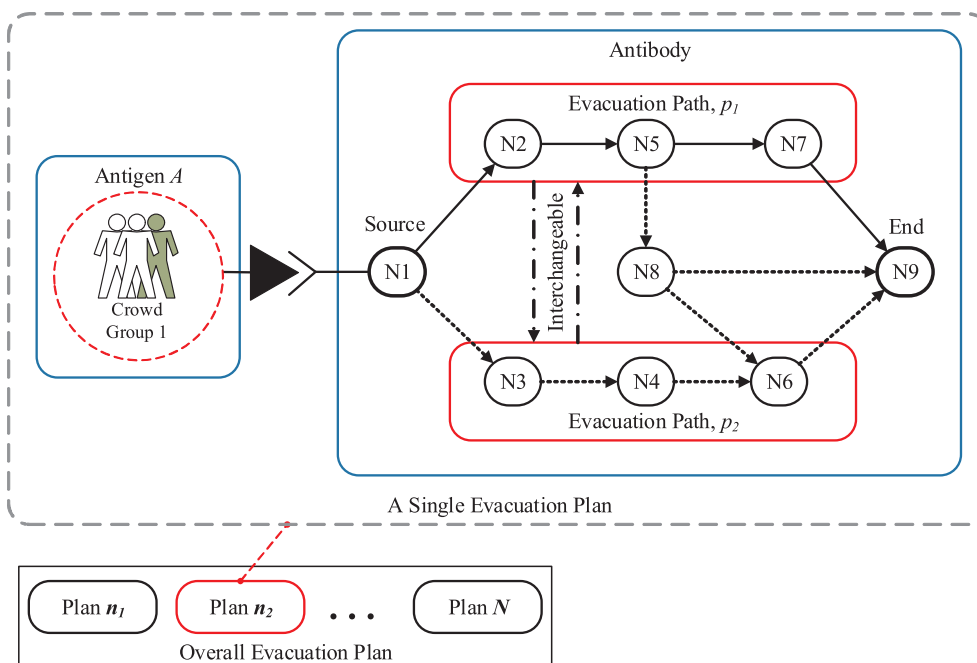


Fig. 1. The dynamic route optimizer of the iEvaP + approach composing dynamic router and route optimization utilizing encoding scheme and altered operators of the proposed AIS.

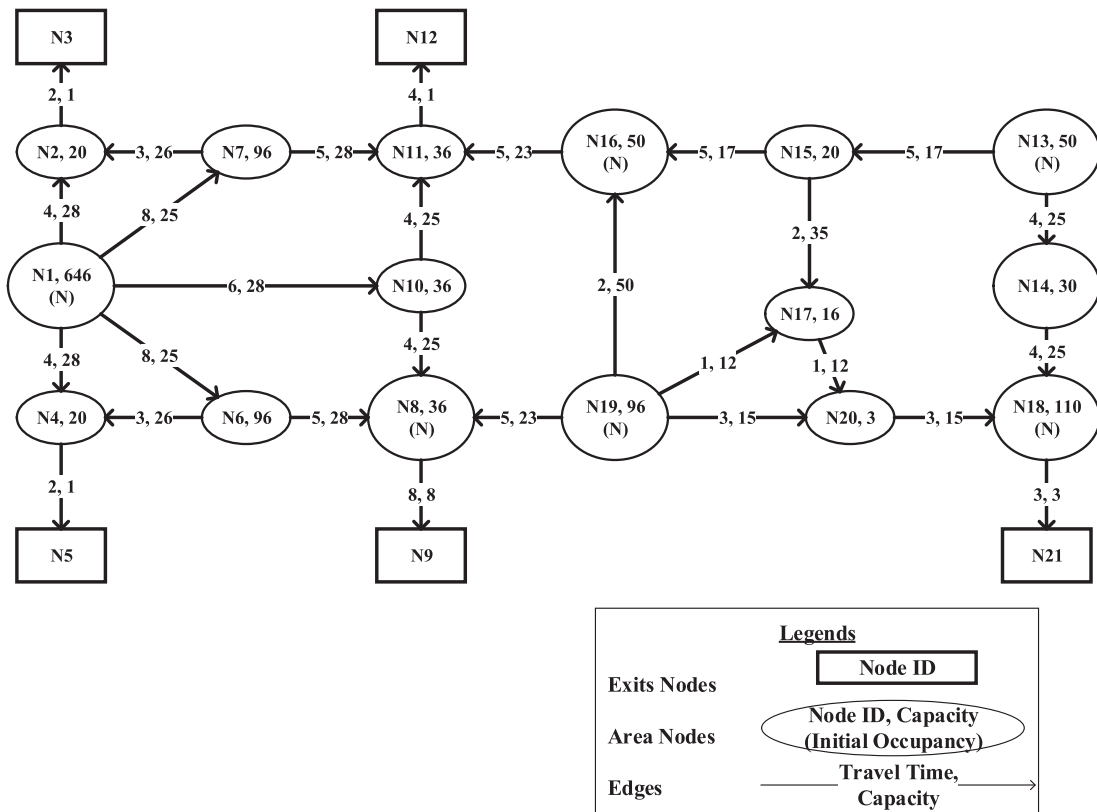


Fig. 2. The logical map of the Bukit Jambul shopping complex (denoted as BJ case study). The data consists of 21 nodes (16 area nodes, 6 source nodes, and 5 exit nodes) and 28 edges.

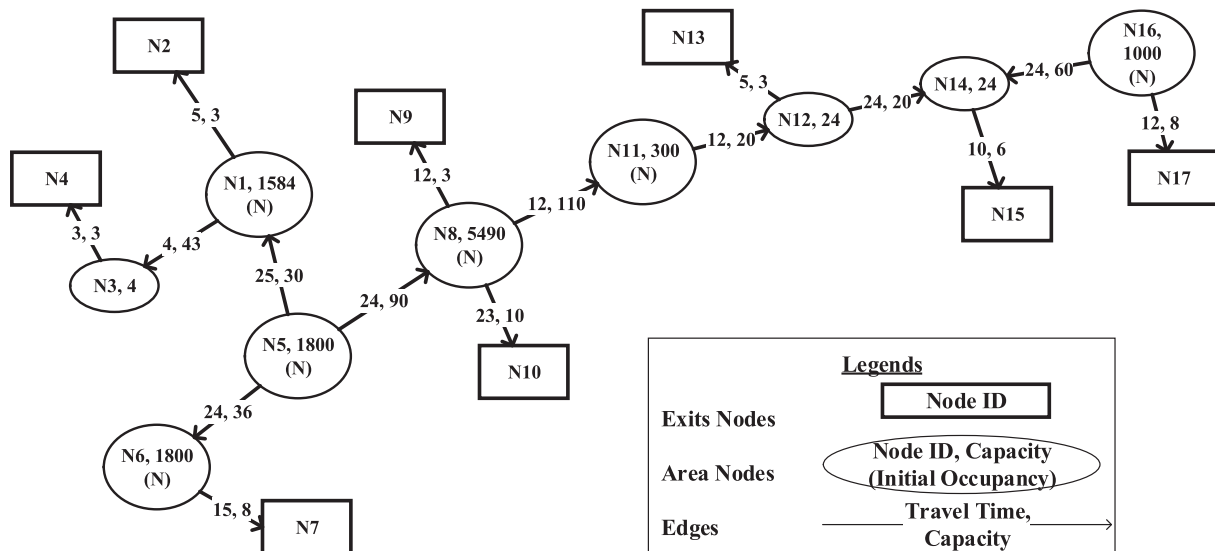


Fig. 3. The logical map of the Queensbay mall complex (denoted as QB case study). The data consists of 17 nodes (9 area nodes, 6 source nodes, and 8 exit nodes) and 16 edges.

originated from broad spectrums of different entities; from human factors (age, size, emotion, relation, etc.), critical event progression (secondary events, smoke density, fire propagation, structural breakdowns, etc.), to engineering factors of the structures (building strength, design layout, material compositions, etc.). Consequently, the scopes and limitations of the study for these particular case studies are defined as follows:

- Critical event is considered existential, although it isn't been reflected directly within the evacuation plan.
- Only group structure and intra-group compliance level are considered in the crowd evacuation model, while others that either

have relation to it (emotion, relationship, genders, etc.) or not (injury, pain, etc.), are neglected.

- The building structure considered for the study is not affected by the pre-disaster or post-disaster impact (structural collapse, severed pathway, etc.).
- Crowd distribution is predetermined and the source to destination pathways are identified in advance in order to simplify the data collection process.

3.4. Computational experiment setup

The computational experiment was designed to identify the impact

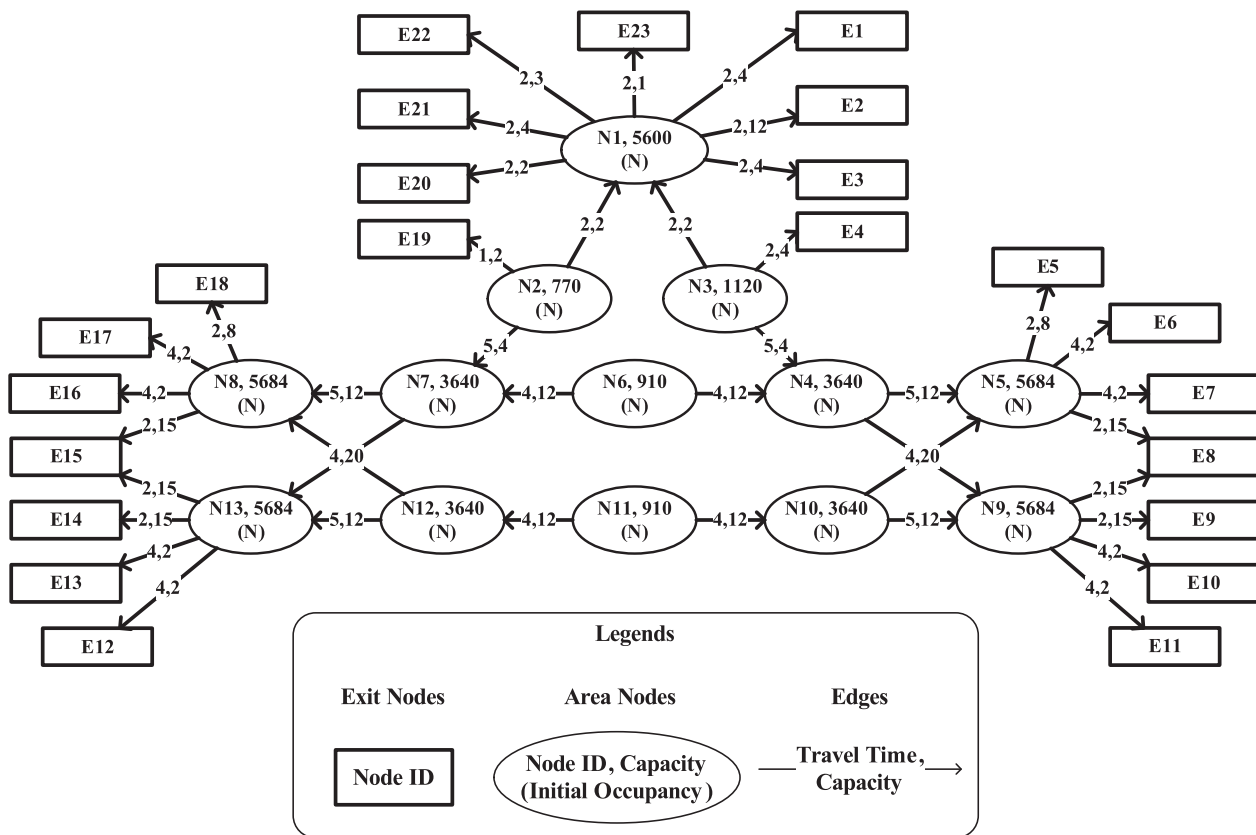


Fig. 4. Logical map of the upper regions of An-Nabawi Mosque (denoted as NM case study) converted from simulation data from [Haron et al. \(2012\)](#). The data consists of 36 nodes (13 area nodes and source node and 23 exit nodes) and 41 edges.

of integrating dynamism in crowd evacuation model (group cohesion) and a route optimizer (AIS) within the proposed iEvaP+ approach using different evacuee sizes. The best performance (minimum of the maximum NCT) out of the 50 samples was taken, as well as their computed average. There were 12 parameter options considered for both iEvaP and iEvaP+ experiments where the options were distinguished by their population sizes (50 and 70) and generation sizes (300 and 500). This produced 4 parameter options for the BJ case study; 8 parameter options for the QB case study; and 2 parameter options for the NM case study. Different total evacuee sizes were considered to emulate realistic crowding of people within an enclosed structure. This corresponded to the number of people that nearly occupied the full capacity of the respective case studies. [Table 3](#) tabulates the parameter settings and parameter options of the aforementioned case studies.

The population sizes (p_{size}), the generation sizes (gen_{size}), and the clone rates (c_{rate}) were chosen on the basis of commonly used parameter settings ([De Castro and Timmis, 2003](#); [Freschi and Repetto, 2005](#); [Aickelin and Greensmith, 2007](#)). The p_{size} may be fixed or incremental ([Timmis et al., 2008](#); [Dasgupta et al., 2011](#)), however the chosen settings were based on justification of preserving diversity due to limited network sizes. Since the AIS algorithm was scarcely adopted for the ERP problem, two gen_{size} values (300 and 500) were adopted to determine the global optimum. The c_{rate} may also be fixed or incremental, however in this study's context, the 0.5 setting is selected because the rate of the solution converging or achieving global optimum can be estimated as well as ensuring solution diversity ([De Castro and Timmis, 2003](#); [Khalid and Yusof, 2014b](#)).

4. Computational results

The summary of the results is given in [Table 4](#). Based on the results,

the best network clearance time (NCT) between the parameter options of similar total evacuee sizes (M), had obtained similar NCT value in most cases, but mainly differentiated by their respective NCT average and standard deviation. For the BJ case study, the best NCT obtained had show no differences between parameter Option 1 ($M = 750$) and Option 2 ($M = 940$). However, small differences were observed on the average and standard deviation of the NCT which implies the reliability of the results obtained. For the QB case study, the best NCT obtained steadily increased with the increase of total evacuee sizes (M). Although parameter Option 4 ($M = 3000$) and Option 5 ($M = 5000$) obtained similar best NCT, the average NCT for parameter Option 5 showed that most of the time the typical NCT value obtained was not similar to the best NCT value. For the NM case study, parameter Option 7b obtained the lowest NCT (and also the best NCT) compared to parameter Option 7a, while their average NCT showed only small differences. However, standard deviation of the NCT implied that minor differences (± 1) of the obtained NCT are observable, most of the time.

The evacuation plan for the BJ case study ([Fig. 5](#)) shows that:

- The exits that were taken the most by the evacuees from various source nodes were N3, N5, N9, and N12. These exits are the exits on the left and middle side of the BJ map.
- Congestion occurred at exit N9 and N12 due to the groups of evacuees coming from source nodes of both the left side (N1 and N8) and right side (N13, N16, and N19) of the BJ map (greyed source nodes in [Fig. 5](#)).
- The number of exit available on the right side is lesser (only exit N21) and the exits support a smaller number of people at a time (a single evacuee at exit N3, N5, and N12; 3 evacuees at exit N21; and 8 evacuees at exit N9).

Meanwhile, the evacuation plan for the QB case study ([Fig. 6](#)) shows

Table 2
Case study's initial evacuee distribution on the source nodes.

BJ Case Study				
Option No.		1a, 1b, 2a, 2b		
Node ID	Initial Crowd Size			Maximum Capacity
	1a, 1b	2a, 2b		
N1	550	640		646
N8	20	35		36
N13	40	45		50
N16	40	45		50
N18	50	90		110
N19	50	85		96

QB Case Study					
Option No.		3a, 3b, 4a, 4b, 5a, 5b, 6a, 6b			
Node ID	Initial Crowd Size				Maximum Capacity
	3a, 3b	4a, 4b	5a, 5b	6a, 6b	
N1	200	400	750	1400	1584
N5	200	250	500	1450	1800
N6	100	400	750	1450	1800
N8	200	1500	2100	4500	5490
N11	50	100	150	250	300
N16	150	350	750	950	1000

NM Case Study				
Option No.		7a, 7b		
Node ID	Initial Crowd Size			Maximum Capacity
	7a, 7b			
N1	5400			5600
N2	700			770
N3	1000			1120
N4	3600			3640
N5	5600			5684
N6	900			910
N7	3600			3640
N8	5600			5684
N9	5600			5684
N10	3600			3640
N11	900			910
N12	3600			3640
N13	5600			5684

that:

- The exits that were taken the most by the evacuees from various source nodes were N2, N4, N7, and N9. These exits are the exits on the left side of the QB map.
- Congestion occurred at the upper left side exits (N2, N4, and N9) and lower left exit (N7) of the QB map (greyed source nodes in Fig. 6) due to groups of evacuees coming from source nodes N1 and N5.
- The exits on the left support a smaller number of people at a time (upper left exit N2, N4, and N9 of QB case study), compared to the right side (exit N15 and N17 of QB case study).

On the other hand, the evacuation plan for the NM case study (Fig. 7) shows that:

- The exits that were taken the most by the evacuees are on the lower left side and lower right side of the NM map (E5 to E11; E12 to E18). The route(s) leading to these exits were shared with the

Table 3
iEvaP+ parameter settings and options.

Parameters		Settings	
Population Size (p_{size})		50, 70	
Generation Size (gen_{size})		500, 300	
Mutation Step Size (m_{steps})		75	
Mutation Rate (m_{rate})		0.01	
Cloning Rate (c_{rate})		0.5	

Parameter Options BJ Case Study			
Option No.	M	p_{size}	gen_{size}
1a	750	50	500
1b	750	70	300
2a	940	50	500
2b	940	70	300

QB Case Study			
Option No.	M	p_{size}	gen_{size}
3a	1000	50	500
3b	1000	70	300
4a	3000	50	500
4b	3000	70	300
5a	5000	50	500
5b	5000	70	300
6a	10,000	50	500
6b	10,000	70	300

NM Case Study			
Option No.	M	p_{size}	gen_{size}
7a	45,700	50	500
7b	45,700	70	300

Table 4
Result summary of the case studies.

Option No.	Exit Number	Result			Congested Exit	
		Best	Average	Standard Deviation		
BJ Case Study	1a	5	32	32.70	0.0005	2
	1b		32	32.62	0.0005	1
	2a		32	32.90	0.0004	1
	2b		32	33.00	0.0004	2
QB Case Study	3a		72	90.08	0.0006	2
	3b		70	89.18	0.0008	1
	4a		91	90.94	0.0003	2
	4b	8	91	91.12	0.0001	3
	5a		91	98.50	0.0004	4
	5b		91	97.66	0.0005	3
	6a		102	102.00	0.0000	4
	6b		102	102.00	0.0000	4
NM Case Study	7a	23	29	31.09	0.7735	14
	7b		28	31.41	0.7593	12

lower middle source nodes (N6 and N11) and upper source nodes (N2 and N3). Both lower middle and upper source nodes meet on both sides of the lower NM map (N7 on left side and N4 on right side).

- Congestion occurred when reaching the lower left and lower right exits (E5 to E11; E12 to E18).
- Both the exit capacity and exit number were satisfactory (7 exits on node N1) on the upper side of the NM map. These situations create balance between congestion occurred on lower left and right side

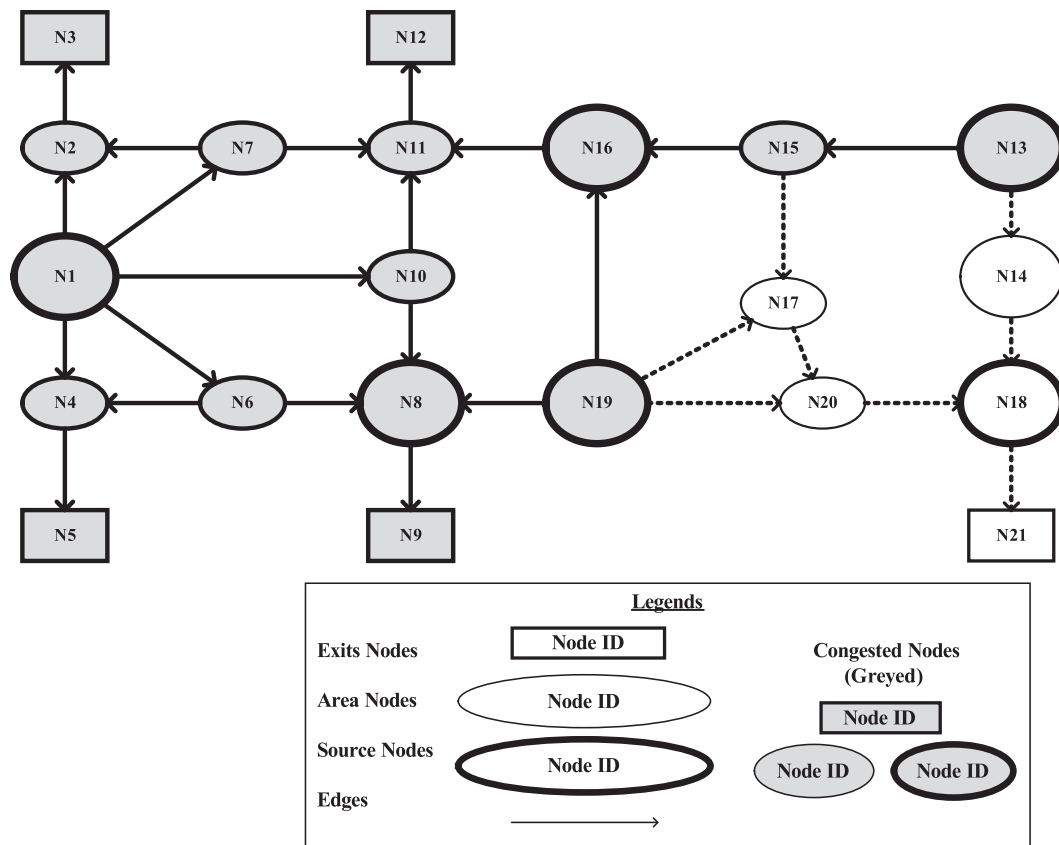


Fig. 5. Logical map of the BJ case study with highlighted routings. The greyed nodes with undashed edges were route(s) taken by most evacuees and indicates occurrence of congestion based on result in Table 4. The dashed edges with white nodes were route(s) least taken by evacuees.

of NM map and route least taken on upper middle NM map.

Both BJ and QB case studies have more exits on the left side of their map compared to the right side. However, the number of exits for the

NM map, compared to the BJ or QB case studies, are nearly three to four times higher, respectively (see [Table 4](#)). As such, the NCT obtained by evacuation plan for the NM case study was much lower than the evacuation plan of BJ or QB case studies.

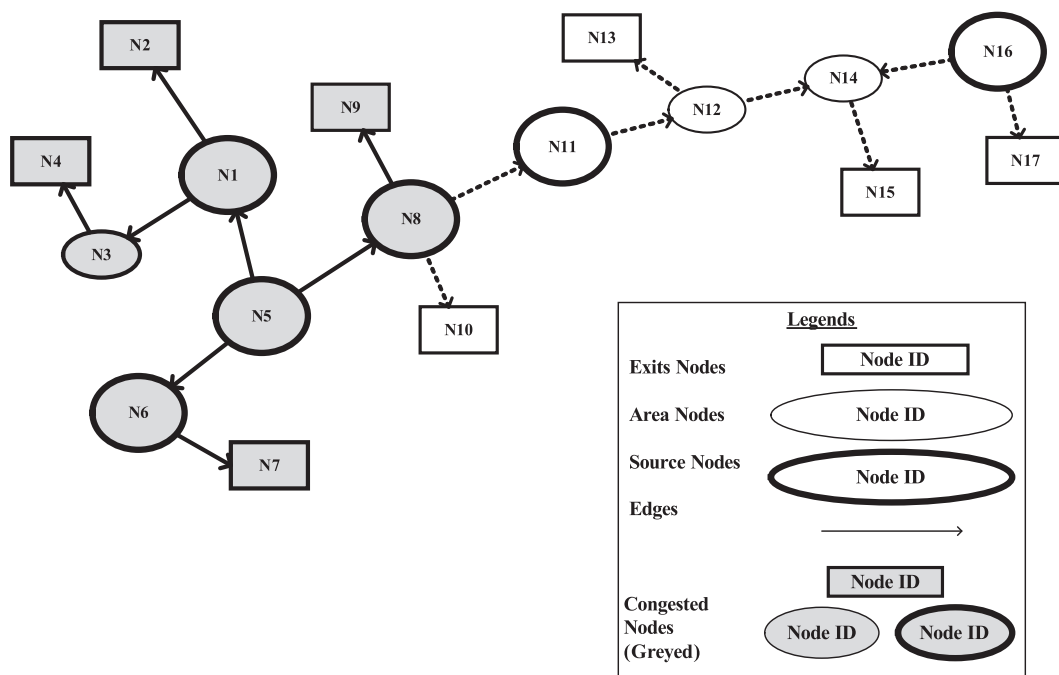


Fig. 6. Logical map of the QB case study with highlighted routings. The greyed nodes with undashed edges were route(s) taken by most evacuees and indicates occurrence of congestion based on result in Table 4. The dashed edges with white nodes were route(s) least taken by evacuees.

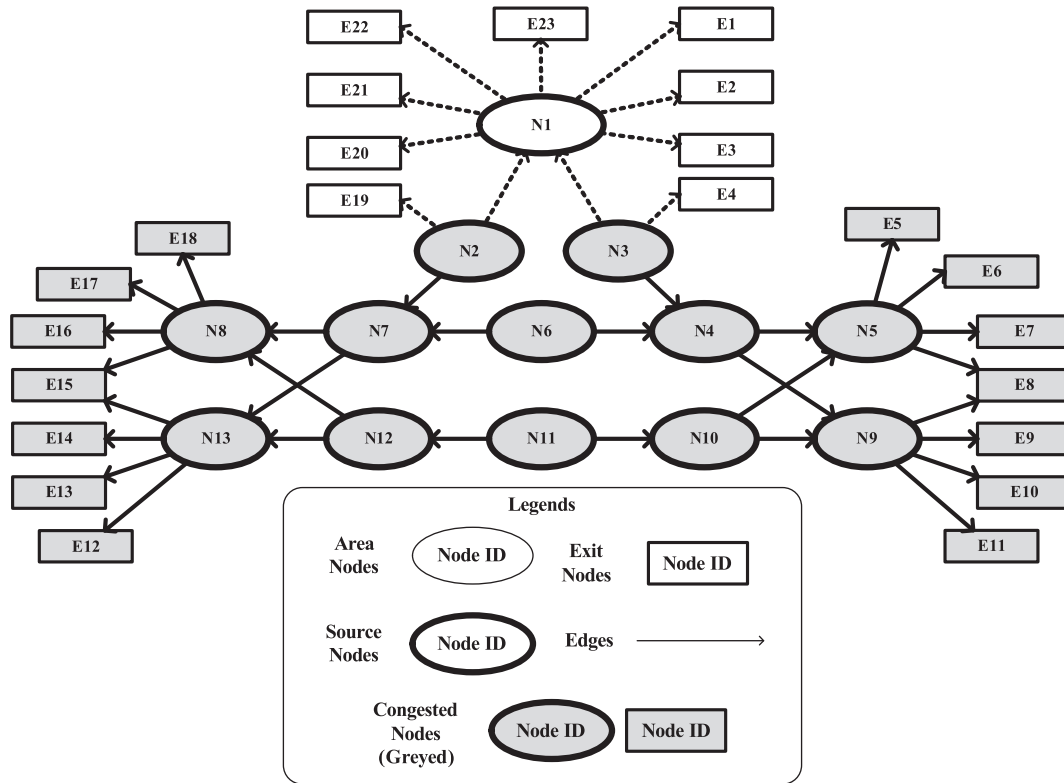


Fig. 7. Logical map of the NM case study with highlighted routings. The greyed nodes with undashed edges were route(s) taken by most evacuees and indicates occurrence of congestion based on result in Table 4. The dashed edges with white nodes were route(s) least taken by evacuees.

5. Analysis and discussions

Mitigating risk is an important aspect of ERP, where regulating the crowd flow and determining the best route selection, prior to a disastrous situation, influences the efficiency of the evacuation plan. Increase in the total evacuee size (M) as well as the group sizing (d_g) has shown to influence the evacuation plan efficiency. In order to identify the correlation between both crowd grouping and intra-group compliance level with their produced evacuation plan performance, the count of group size (d_g) and the count of compliance rate of groups ($rate_g$) were recorded in Table 5. The table also shows the group size composition and the intra-group compliance level of each evacuee from the total evacuee sizes of the respective parameter options. Note that the recorded count of group size (d_g) and the count of compliance rate of groups ($rate_g$) were based on the solutions that obtained the best NCT in Table 4. The difference between the cooperative $rate_g$ (≥ 60) and the rebellious $rate_g$ (<60) were calculated using Eq. (3), where r_a is the cooperative $rate_g$, and r_b is rebellious $rate_g$.

$$\text{Difference} = \left\{ \frac{r_a - r_b}{(r_a + r_b)/2} \right\} * 100. \quad (3)$$

An increase in total evacuee size induces a gradual increase of best NCT as well as the average NCT (refer Table 4). These phenomena were caused by the varying $rate_g$ and d_g . A high total count of ideal d_g (refer to the work by Carron and Brawley (2000), Hackman (2012), Moreland et al. (2013) on ideal group structure) corroborates with the NCT improvement of the evacuation plan, especially the one with cooperative $rate_g$. The difference between the cooperative $rate_g$ (≥ 60) and the rebellious $rate_g$ (<60) is greater than 50% in Option 1b, 2a, 3b, 4a, 5b, 6a, and 7b. A higher percentage of (cooperative) $rate_g$ emulates a group with cooperating members which consequently improves the group's flow rate, thus achieving a better NCT compared to a group with low percentage of (rebellious) $rate_g$.

In addition, ideal group sizes ($2 \leq d_g \leq 5$) as well as cooperative

intra-group compliance levels ($60\% \leq rate_g \leq 100\%$) are the reasons for enabling a smooth crowd flow rate, where better NCT values are obtained for parameter Options 1b, 2a, 3b, 4a, 5b, 6a, and 7b. A higher value of the total evacuee size (M) increases the probability of having larger group sizes ($5 < d_g \leq 9$). This increases the possible inter-group competition. The capacity constraint of the network layout would hinder crowd flow rate, especially larger group sizes. This incurs additional delays, thus increasing the NCT values. This situation imitates real-world blocking and congestion, where one group has to compete with another group for available route capacity. This induces a “waiting” time that would renders emergency evacuation plan inefficient (Qiu and Hu, 2010).

Increasing the value of the total evacuee size (M) increases the possibility of having lower (rebellious) compliance levels ($10\% \leq rate_g < 60\%$). This increases the possible intra-group competition. As intra-group competition causes delays, these additional delays would also increase the NCT. This situation imitates pushing and pulling between individuals within the same group, during an evacuation, in order to protect oneself or others and expressions of panic and stress toward advent danger (Moreland et al., 2013). Although these real-world crowd behaviors are not incorporated exclusively, the main foundations of these behaviors can be implicitly modeled. Nevertheless, every respective case study have indicated the associated effects on the efficiency of an evacuation plan when total evacuee size (M) increases.

Furthermore, group distribution for the BJ case study is smaller (total group count up to 172 groups) compared to the group distribution for the QB case study (total group count up to 373 groups), while the QB case study is smaller compared to NM case study (total group count up to 1021 groups). By optimizing the route taken by these groups, obtaining the best NCT values of the overall evacuation plan was possible even when encountering capacity constraints from the evacuation network. By assigning suitable routes for each group, the value of NCT can be reduced regardless of the actual flow rate of the

Table 5Group size (d_g) distribution and intra-group compliance rate ($rate_g$) distribution of iEvaP+ based on the parameter options.

Options	Total Evacuee Size (M)	d_g count				$rate_g$ count		
		$d_g = 1$	$2 \leq d_g \leq 5$ (ideal)	$d_g > 5$	Total	$60 \leq rate_g \leq 100$ (cooperative)	$rate_g < 60$ (rebellious)	Difference (%)
<i>BJ Case Study</i>								
1a	750	11	26	67	104	453	297	41.60
1b		31	71	30	132	537	213	86.40
2a	940	92	129	43	264	766	174	125.96
2b		25	30	118	173	532	408	26.38
<i>QB Case Study</i>								
3a	1000	11	37	113	161	527	473	10.80
3b		35	152	95	282	695	305	78.00
4a	3000	45	121	82	248	2183	817	91.07
4b		33	149	187	369	1484	1516	2.13
5a	5000	44	133	212	289	3015	1985	41.20
5b		24	201	75	300	3458	1542	76.64
6a	10,000	45	226	130	401	7432	2568	97.28
6b		51	139	234	424	5423	4577	16.92
<i>NM Case Study</i>								
7a	45,700	246	432	589	1267	20,234	25,466	22.90
7b		224	568	443	1235	34,524	11,176	102.18

evacuees throughout the assigned route. Taking into account the NCT value obtained from all the routes of the groups, a small reduction of the flow rate could potentially lead to significant improvements of the overall performance of an evacuation plan.

On the basis of inter-group and intra-group competition, as well as the group distribution in general, it can be concluded that the crowd dynamic model (specifically, crowd group cohesion) has significant impact on the crowd evacuation plan. Without group cohesion as part of the crowd model, the evacuation plan may be better in terms of NCT, but it causes the evacuation plan to be infeasible due to unforeseen delays caused by the crowd when an actual disastrous situation occurs. Therefore, it is vital to consider crowd dynamism to mitigate actual risks and potentially minimize loss of life. However, only adopting a suitable crowd model is sufficient. Identifying the impact of utilized exits and their properties are also important in producing an optimum evacuation plan.

The variation of result patterns observed from the first two case studies (BJ and QB case study) are mainly caused by their structural properties, such as limitation in exit numbers and narrow exit(s). As discussed in the previous section, congestion occurred at up to two exits and four exits of the evacuation plan for the BJ and QB case study, respectively (Table 4). Also, congestion occurred at up to 14 exits of the evacuation plan for the NM case study. By correlating the NCT obtained from each option of the respective case study (Fig. 8(a)), the best NCT obtained, with respect to their option number (1b, 2a, 3b, 4a, 5b, 7b), showed fewer exits that are congested (cf. Table 4). Although, for example, Option 1a and 1b obtained similar best NCT, Option 1b is better (Fig. 8(a)) because fewer exits were congested. This situation also applies for Option 2a, 3b, 4a, 5b, and 7b.

The exits of each case study also supports different numbers of evacuees at a time. The BJ, QB and NM case studies have average exit capacities of 2.8, 5.5, and 6.9, respectively (each was computed by summing up all the exit capacities and then, dividing them by the total exit number). Also, the total exits of each case study are 5, 8, and 23, respectively (see Table 4). To analyze further, the average exit capacity is divided with the total exit number, so that their ratio (denoted as exit rate ratio) can be obtained (Fig. 8(b)). This exit rate ratio can be used to indicate the relationship between the average exit capacity and exit numbers. The higher the average exit capacity (the wider the exit), the

lower the exit numbers; and vice versa. The logical map of the NM case study gave an exit rate ratio of 0.3, BJ case study gave an exit rate ratio of 0.56, and QB case study gave an exit rate ratio of 0.69.

Based on the results of each case study (see Table 4), it can be highlighted that the NCT obtained by evacuation plan for NM case study is the lowest, followed by the BJ case study and finally the QB case study. This pattern is reflected by the aforementioned exit rate ratio (Fig. 8(b)). The logical map of the NM case study has a high number of exits although its average exit capacity is only 6.9 (wider exit compared to other case studies). This means that the NM case study has a large number of exits with satisfactory exit width. Since the exit capacity limitation is addressed by the high exit numbers; a low NCT value was obtained. On the other hand, the logical map of QB case study has a low number of exits (8 total exits) but its average exit capacity is 5.5 (quite close to NM case study but narrower). This means QB case study has a high number of narrow exits. The logical map of the BJ case study has a fairly balanced ratio between total exit number and average exit capacity (exit rate ratio of 0.56). This means that the BJ case study has a small number of narrow exits. With the increase in total evacuee sizes M , the evacuation plan for the QB case study has obtained high NCT value. This is because the exit capacity limitation hinders the evacuation flow, as well as there being a limited number of available exits. However, the BJ case study handles smaller total evacuee sizes M ; thus resulting in a low NCT that is still higher than the one produced by the evacuation plan of the NM case study.

Based on the computed exit rate ratio and the results obtained (Table 4), both logical maps of the BJ and QB case study share similarities in terms of having insufficient or narrow exits. Both case studies require more exits for accessibility to increase the number of evacuees their exits can support at any given time (retain the current average exit capacity, but increase the total exit numbers; or vice versa). In other words, a small number of wide exits or a large number of narrow exits. However, the NM case study can no longer improve its logical map by any adjustments (increasing either average exit capacity or total exit numbers) due to concerns on security and/or costs of the respective case study structure (exit too wide or too many exits). Nevertheless, all case studies must have balance between wider exits and high exit numbers in order to improve the rate of evacuation. Therefore, it can be concluded that exit properties (exit width and number) significantly

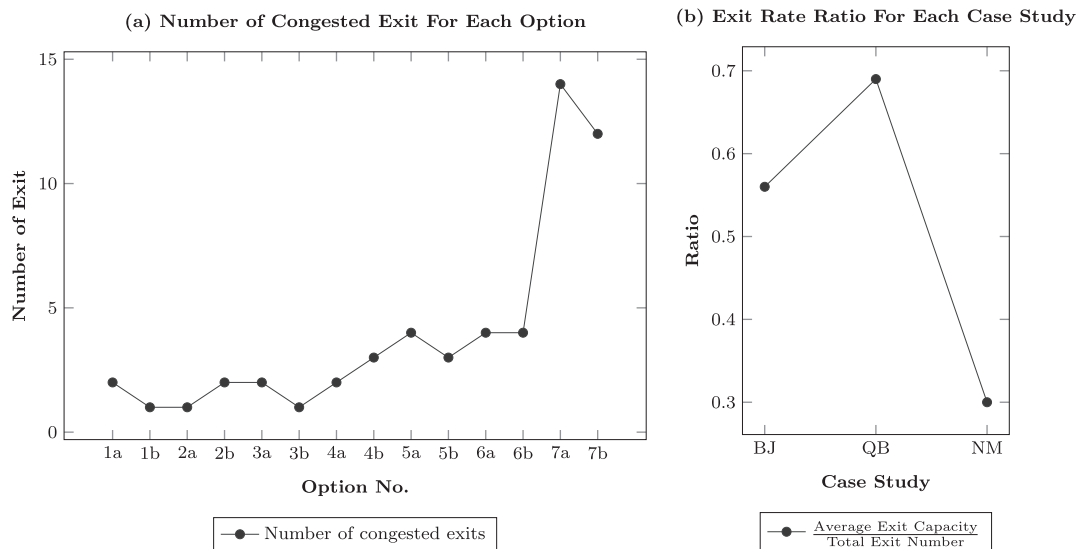


Fig. 8. (a) Correlation of congested exits against the option numbers, where options that obtained best NCT (see Table 4) are the one that have less congested exits; (b) Computed exit rate ratio based on the average exit capacity divided by the total exit numbers, where case study with low exit rate ratio gave better NCT (see Table 4) compared to high exit rate ratio.

affects the ability of a crowd to evacuate.

Although adopting a suitable crowd model and considering exit properties are important, utilizing an appropriate route assignment algorithm is also important in producing an optimum evacuation plan. In the QB case study, the performance of the route assignment algorithm (AIS algorithm) may be affected due to limited exit path's capacity. Although the AIS algorithm's operators can successfully capture the logical structure of the QB case study (which is limited in nodes and edge numbers), the operators of the optimization algorithm may have little impact in the improvisation process. This situation implies that algorithmic searching may have little impact on evacuation plan improvement. This case is in contrast to the BJ case study, where the edge branching from a higher node has a higher impact when the route assignment is improvised. As such, this will potentially affect the overall evacuation plan to be come much better or worst. A similar situation applies to the NM case study, where edge branching from a node is higher than the one in BJ or QB case study. This enables more route improvisation possibilities to the respective optimization algorithm, thus creating more route flexibility.

As meta-heuristic approaches produce results by means of approximation, the exact outcome is impossible to predict or estimate (Talbi, 2009). Based on such notion, the cause of the differences between the best NCT values of Options *a* and the best NCT values of Options *b* are likely from the algorithmic operator (c_{rate}) and size of the population (p_{size}) adopted. Since Option *b* utilizes a populations of 70, compared to 50 populations of Option *a*, the diversity of the solution's population may be compromised from similar c_{rate} value used. The c_{rate} value is an operator that involves cloning the solution (0.5 or 50% chance) and m_{rate} and m_{steps} are fixed parameters that reflect the improvisation process of the solution. These parameter selections of the operators could results in either better or worse solutions. Based on the presented results, both Options *a* or *b* may be adopted for different total evacuee sizes (M) but from differences in the obtained results can only occur when there are differences in the layout of the evacuation network (edge branching and exit numbers).

6. Concluding remarks

The iEvaP+ approach has considered the AIS algorithm for its dynamic route optimizer while capturing group cohesion as its dynamic crowd model, encompassing group structure and intra-group compliance levels in order to elicit an optimum evacuation plan. The iEvaP

+ approach is applied to three actual case studies. These three case studies were collected from the layout of Bukit Jambul shopping complex, Queensbay mall, and the An-Nabawi mosque. The goals of this study are twofold; identifying the impact of incorporating dynamism in solving the ERP problem and identifying the relationship between the exit properties (exit width and number) and the ability of the crowd to evacuate. The results show that the obtained evacuation plan is directly affected by dynamism when the optimum NCT is obtained.

In reality, a crowd moves in groups and has varying affecting factors (such as compliance, speed, and direction), in which, differences in group structure and their respective compliance level may affect the outcome of a preplanned emergency evacuation. Additionally, even the best designed logical map (i.e., optimum exit rate ratio) cannot guarantee the ability of the evacuee to egress with relative ease. The proposed iEvaP+ approach acts as a medium for a robust and efficient evacuation planning and management solution. Also, the immediate bottlenecks and possible occurrence of crowding can be avoided. This is valuable in mitigating risks by preparing for an actual occurrence of emergency.

The possible future works are outlined as follow:

- The proposed optimization algorithm (AIS algorithm) can be applied to similarly constrained optimization problems, particularly in allocation and design, which includes but is not limited to, the areas of the logistical management, ambulance dispatching, and shelter allocation of the ERP problem domain. In addition, this study only considers certain constraints with the assumption of evacuees are independent of other external constraints such as local authority, leaders, signage, and visibility level. The extension of this study should cover these constraints.
- This study measures the exit properties and their impact on the produced evacuation plan, which highlights the importance of properly designed exits of a structure. Further improvement can be made by optimizing the exit rate ratio for certain evacuee sizes and logical structures. Additionally, the exit rate ratio can be potentially applied to domain problems that involve the safety of an enclosed structure; such as design optimization and related field.
- This study incorporates the dynamic crowd model which uniquely focused on group cohesiveness, while neglecting other psychological and physiological human factors. Future work could consider integrating more complex human factors such as emotions (e.g. fear, anxiety, etc.), physical attributes (e.g. age, cultural preference,

injury, etc.), and others (e.g. disabled person, children, etc.). In addition, the occurrence of the disastrous situations can also be incorporated to simulate actual impact it has towards the survivability and safety of the crowd.

- (d) This study incorporates a time-based performance, namely network clearance time, as the performance measure of the emergency evacuation plan. However, no other performance measures have been considered, despite the fact that potentially new performance measures, such as total fatalities, total cost, estimated affected area m^2 , etc., have been introduced. A multi-objective optimization or multi-attribute optimization technique can be considered in the future where the process treats each objective separately, while simultaneously optimizing one or more conflicting objectives with respect to certain constraints.
- (e) The current approach can be improved by incorporating the latest modelling technology such as multi agent system, and visualization technique such as 3-dimensional visualization of the building structure and the evacuees. In addition, a finer evacuation simulation can be analyzed and visualized to aid decision support for mitigating risks in the context of crowd safety and crowd survivability.

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